

Project Title: Classifying Credit Scores

1. Dataset Overview and Preprocessing

The dataset consisted of 50,000 records related to customers' financial, demographic, and behavioral characteristics. The goal was to classify individuals into credit score categories based on these features.

Preprocessing steps included:

- Dropping non-informative or redundant columns: ID, Customer_ID, Name, SSN, Month
- Handling missing values:
 - Numerical columns were filled with median values.
 - Categorical columns were filled with the mode or with a placeholder like "Unknown."
- Encoding categorical variables:
 - Label encoding was applied to ordinal features like Payment_Behaviour, Credit_Mix, and Occupation.
 - Multi-label features like Type_of_Loan were expanded using one-hot encoding.
- Outliers were capped using the IQR method to reduce their influence without dropping records.
- The target variable Credit_Mix was encoded as:
 - Bad: 0, Standard: 1, Good: 2

2. Exploratory Data Analysis (EDA)

- The distribution of the target variable Credit_Mix was:
 - Standard: 56.4%

- Good: 24.5%
 - Bad: 19.1%
 - Although slightly imbalanced, the distribution did not require synthetic oversampling like SMOTE.
 - Important features based on feature importance and SHAP:
 - Interest_Rate
 - Payment_of_Min_Amount
 - Outstanding_Debt
 - Num_Bank_Accounts
 - Num_of_Delayed_Payment
 - Visualizations included count plots, box plots, correlation heatmaps, and SHAP summary plots to understand feature relevance and interaction.
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3. Model Building and Evaluation

The dataset was split into training and testing sets using stratified sampling to preserve the class distribution.

The following models were trained and evaluated:

- Logistic Regression
- Random Forest
- XGBoost
- Gradient Boosting Classifier
- AdaBoost
- LightGBM

- CatBoost

Performance Summary:

Model	Accuracy	Strengths	Limitations
Logistic Regression	0.72	Fast, interpretable, baseline model	Weak with non-linear relationships
Random Forest	0.86	High accuracy, robust, interpretable features	Slightly less interpretability in decision paths
XGBoost	0.85	Accurate, handles imbalance	Requires careful tuning
Gradient Boosting	0.84	Balanced metrics, effective learner	Sensitive to overfitting, slower training
AdaBoost	0.77	Simple, fair class 2 performance	Lower accuracy, affected by noise
LightGBM	0.85	Fast, efficient, good with large data	May be sensitive to feature formatting
CatBoost	0.85	Strong with categorical features	Higher memory usage, longer training

4. Hyperparameter Tuning

Random Forest was tuned using GridSearchCV with the following parameter grid:

- n_estimators: [100, 200]
- max_depth: [5, 10]
- min_samples_split: [2, 5]

Best parameters found:

- max_depth: 10
- min_samples_split: 5

- n_estimators: 100

Post-tuning accuracy: 0.8443

5. Model Interpretability

SHAP values and feature importance analysis showed:

- The most impactful features were Interest_Rate, Payment_of_Min_Amount, and Outstanding_Debt.
- These insights help in understanding how decisions are made and can assist in feature auditing and trust-building in deployment.

6. Conclusion and Recommendations

- Random Forest performed the best overall with high accuracy and balanced precision and recall.
- Interpretability techniques confirmed that key financial behavior features were driving predictions.
- The preprocessing pipeline was robust and scalable for new data.
- The model is ready for deployment with additional monitoring for class balance and feature drift.

Recommended next steps include:

- Deploying the Random Forest model with monitoring
- Implementing a feedback loop to improve data quality
- Using SHAP in production for instance-level explanations